

Location and contract pricing of distributed generation using a genetic algorithm

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ABSTRACT

This paper presents an approach based on a specialized genetic algorithm (GA) to determine the location and contract pricing of dispatchable distributed generation (DG) units in distribution systems. The proposed approach is based on a nonlinear bilevel programming framework that involves the interests of two different agents: the DG owner who procures the maximization of the profits obtained from the energy sales, and the Distribution Company (DisCo), which procures the minimization of the payments incurred in attending the forecasted demand. To meet the forecasted demand the DisCo can purchase energy either from the wholesale energy market or from the DG units within its network. The proposed GA determines both the location and contract pricing of the DG units that would render maximum profits to the DG owner, subject to the minimization of payments procured by the DisCo. To show the effectiveness of the proposed approach, several tests were carried out on a modified IEEE 34-bus and 85-bus distribution networks.

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1. Introduction

Distributed generation can be understood as the production of electricity by small generators located in the distribution network or near the loads they are attending [1]. In the past few years the electric industry has witnessed an increased interest in DG. This interest has been motivated by several factors such as recent advances in generation technologies that have rendered generation units smaller and feasible, an increasing awareness of environmental issues, and the need for more flexible electric systems.

The technical and economic impacts of DG have been widely studied in the specialized literature [2–7]. As most of the potential benefits of DG depend on the size and location of the new generating units, their optimal location and sizing has received special attention. In this regard, there are several methodologies that have been proposed in the literature to optimally size and allocate DG. These methodologies include analytical approaches [8,9], classical optimization [10–12] and metaheuristics [13–18].

Ref. [8] presents analytical approaches for optimal placement of DG in distribution systems considering uniformly, centrally, and increasingly distributed loads. In [9] an analytical expression and a methodology based on the exact loss formula are proposed in order to determine the size and location of DG that would minimize power losses. In [10] a methodology based on classical optimization is proposed to determine the optimal placement

and penetration level of DG with the objective of minimizing costs and power losses. In [11] the locational marginal prices (Imps) along with the consumer payment are used to identify the candidate buses where to allocate the DG. In [12] an optimization model including binary decision variables is proposed to solve the distribution system planning problem including the size and location of new DG units.

Several metaheuristic approaches have also been applied to the problem of finding the optimal location of DG units. In [13] a multi-objective evolutionary algorithm is proposed to maximize the benefits of DG by limiting the deterioration of the network performance due to DG units not connected in optimal locations. In [14] a GA is presented to determine the allocation and sizing of DG considering nodal pricing for profit, loss reduction and voltage improvement. In [15] a Tabu Search metaheuristic is proposed to find the optimal location of DG units that would minimize power losses. In [16] this technique is also applied using a multi-objective optimization approach. In [17] a Particle Swarm optimization technique for the optimal allocation of DG is presented, being the objective function the minimization of power losses. In [18] this technique is also applied to improve voltage profile and reduce total harmonic distortion as well as power losses. The main difference between the aforementioned methodologies and the proposed GA consists in a bilevel approach to the problem that considers a different objective function.

A bilevel programming problem (BLPP) is a hierarchical optimization problem consisting of two levels. Each level corresponds to an agent with an objective function and subject to a set of constraints. The upper level agent is known as the leader and is

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Nomenclature

Indexes

n, m	bus indexes
j	distributed generation unit index
l_{mn}	index of line connecting nodes m, n

Parameters

$\rho_{SE}(t)$	wholesale energy price at the substation in period t (€/MW h)
Δ_t	length of the time interval t (h)
S_{lmn}^{Max}	maximum apparent power limit in the line connecting nodes m, n (MVA)
V_n^{Max}	maximum voltage limit in bus n (V)
V_n^{Min}	minimum voltage limit in bus n (V)
P_{DGj}^{Max}	maximum active power limit of DG unit j (MW)
P_{DGj}^{Min}	minimum active power limit of DG unit j (MW)
P_{SE}^{Max}	maximum active power limit of the substation (MW)
P_{SE}^{Min}	minimum active power limit of the substation (MW)
Q_{SE}^{Max}	maximum reactive power limit of the substation (MVAR)
Q_{SE}^{Min}	minimum reactive power limit of the substation (MVAR)
Q_{DGj}^{Max}	maximum reactive power limit of DG unit j (MVAR)
Q_{DGj}^{Min}	minimum reactive power limit of DG unit j (MVAR)
c_{DGj}	production cost of DG unit j (€/MW h)
$P_{Dn}(t)$	active power demand in bus n in period t (MW)
$Q_{Dn}(t)$	reactive power demand in bus n in period t (MVA)

g_{mn}	real part of element m, n of the admittance matrix (mho)
b_{mn}	imaginary part of element m, n of the admittance matrix (mho)
N_{DG}	number of DG units

Variables

$P_{SE}(t)$	active power supplied by the substation in period t (MW)
$Q_{SE}(t)$	reactive power supplied by the substation in period t (MVA)
$P_{DGj}(t)$	active power supplied by the DG unit j in period t (MW)
$Q_{DGj}(t)$	reactive power supplied by the DG unit j in period t (MVA)
$V_n(t)$	voltage magnitude of node n in period t (V)
θ_{mn}	angle between nodes m, n
λ_{DGj}	contract price of DG unit j (€/MW h)
μ_{nj}	binary decision variable indicating the allocation of DG unit j in bus n
P_{lmn}	real power in line connecting nodes m, n (MW)
Q_{lmn}	reactive power in line connecting nodes m, n (MVar)
S_{lmn}	apparent power in line connecting nodes m, n (MVA)

Sets

J	set of indexes of DG units
T	set of time intervals
N	set of indexes of network nodes

subject to the optimization problem of the lower level agent known as the follower. In this case, the two decision-making agents are the DG owner (leader) and the DisCo (follower). On one hand, given a fixed number of DG units with a predefined capacity, the DG owner pretends to maximize the profits that would be obtained from the energy sold to the DisCo. For this, he must decide the most suitable locations and contract price offers for each DG unit. On the other hand, the DisCo procures the minimization of the payments incurred in attending the forecasted demand. For this, the DisCo must decide the quantity of energy to be purchased from the DG units and from the wholesale energy market through the substation.

To decide the amount of energy to be purchased from each supplier, the DisCo must take into account not only energy prices, but also the impact that this energy has in its network. An effective way to consider these two aspects is by means of an optimal dispatch based on an AC optimal power flow.

In a conventional wholesale electricity market generating agents decide over their price and quantity bids, and are paid at the market clearing price. However, in the market structure envisaged in this paper the DG owner decides over the contract prices, while the DisCo determines the quantity of energy to be purchased, consequently, only dispatchable DG technologies are considered in the model. Furthermore, a single DG owner who might have several DG units is considered. For each unit, the DG owner is interested in finding the location and contract price that would maximize total profits.

The main feature of a BLPP is that the decisions made by the upper level agent (the DG owner) must anticipate the reactions of the lower level agent (the DisCo). In this sense, a bilevel programming problem is equivalent to a single-round Stackelberg game [19]. The players of this game are the leader and the follower. The leader makes its move first anticipating the reaction of the follower, and then the follower makes its move reacting to the strat-

egy adopted by the leader. In this case the leader is the DG owner who must decide over the location and contract prices of his units, and the follower is the DisCo which reacts to the offer of the DG owner buying more or less energy. An interesting fact is that the DG owner knows the DisCo will decide over the energy to be purchased based on an optimization procedure. Then, it anticipates the reaction of the DisCo considering such optimization procedure as one of the constraints of its own optimization problem. This relationship is depicted in Fig. 1.

Due to its hierarchical structure even a BLPP with linear upper and lower level optimization problems (known as linear BLPP) is NP-hard [20]. Therefore, a nonlinear BLPP that includes integer variables, as the one in this paper, is a more complex and challenging problem. These types of problems can be better solved using metaheuristics than conventional optimization procedures [21].

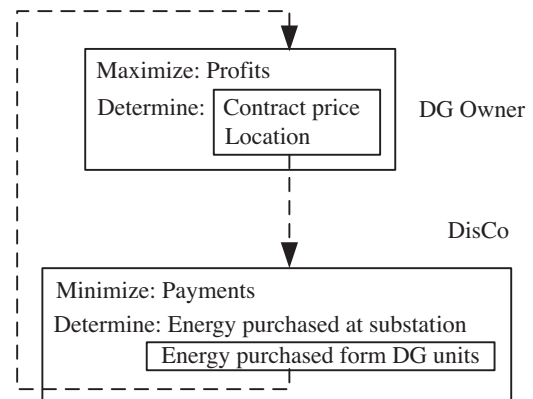


Fig. 1. Bilevel programming diagram.

The proposed approach for location and contract pricing of DG units is based on the assumption that DG units are either controlled or respond to control orders given by the DisCo. While it is a fact that small consumers connected at low voltage might not be interested in engaging in such a business scheme with the DisCo, bigger consumers, connected at medium voltage levels might be. This last case is common in Brazil where DG units powered by biomass (sugarcane bagasse) or biogas are planned not only to attend the local demand, but also to export power to the main grid. In the latter case, the extra energy is negotiated through long-term bilateral contracts. Consequently, the DG does not mind being scheduled by the DisCo, provided that the amount of energy and price already negotiated does not change. In Great Britain the DG can sell its energy in the market or to the Distribution Operator through an aggregator (an agent that might represent a set of small generators) and enter into a bilateral contract. In the latter case such contract might set out features regarding the operation of the DG which would influence how and when it is operated. In this way, the utility, through the aggregator, has some sort of control over the DG. A similar scheme can be applied in Spain. Current regulation does not allow the DisCo to trade directly with DG; however, it is possible to establish the provision of energy, and other services, through a retailer. In the latter case the DG might be flexible regarding its operation, for the sake of profit maximization.

A first attempt restricted to the optimal contract pricing of DG under a bilevel programming framework has been proposed by the authors in [22]. In this case the BLPP is turned into a standard single-level non linear optimization problem and solved using commercially available software. However, the model proposed in [22] presents two main drawbacks: (i) it is based on approximations of the power flow equations that neglects the effects of reactive power and voltage angle differences between buses, and (ii) the lower-level problem is substituted by its Karush–Kuhn–Tucker optimality conditions, which, for a non linear programming problem, constitute a necessary but not sufficient condition of optimality. In contrast, the model presented in this paper does not consider power flow approximations, and the lower-level problem is explicitly solved in each iteration. Furthermore, besides contract pricing, it also considers the optimal location of DG units. Consequently, the proposed model is a two-level mixed integer non linear optimization problem, considerably more complex than the model presented in [22].

The contributions of this paper are fourfold.

- (1) It provides a framework for a DG owner based on a bilevel structure, which allows selecting its optimal contract price and location.
- (2) No approximations on the power flow equations are needed, as in previous works that rely on conventional branch-and-cut solvers.
- (3) The proposed approach allows the DG owner to anticipate the reaction of the DisCo.
- (4) The interests of both agents, the DisCo and DG owner, are explicitly considered in the model.

The remainder of this paper is organized as follows: Section 2 describes the mathematical formulation of the nonlinear BLPP, Section 3 presents the proposed GA to solve the BLPP, Section 4 presents some tests and results, and Section 5 presents the conclusions and final remarks.

2. Problem formulation

The mathematical formulation of the nonlinear BLPP is provided in (1)–(13). The upper level problem given by (1)–(3) refers to the

DG owner. Eq. (1) corresponds to the maximization of his profits, which can be calculated as the difference between the contract price and the production cost multiplied by the energy sold. Eq. (2) establishes the binary condition of μ_{nj} , if is equal to one indicates that the DG unit j is allocated at bus n . In this case there are N_{DG} units with a predefined capacity. Eq. (3) establishes that the same DG unit is not allocated in more than one bus. The decisions in the upper level problem consist in determining the contract prices and locations of the DG units that would maximize the profits of the DG owner. These decisions are parameters in the lower level problem.

The lower level problem is given by (4)–(13) and consists in an optimal dispatch that must be solved by the DisCo. Eq. (4) corresponds to the objective function, which is the minimization of the payments incurred in attending the forecasted demand. Eqs. (5) and (6) correspond to the power balance equations at the substation. Eqs. (7) and (8) correspond to the power balance equations at the nodes different from the substation. Eqs. (9) and (10) correspond to the active and reactive power limits of the substation, respectively. Eqs. (11) and (12) correspond to the active and reactive power limits of the DG units, respectively. Finally, Eqs. (13) and (14) correspond to the limits of the voltage magnitudes of the nodes and the power flows in all the lines of the network, respectively. The energy purchased by the DisCo must match the forecasted demand plus the energy losses; consequently, minimization of active power losses is implicit in (4).

$$\text{Max}_{\lambda_{DGj}, \mu_{nj}} \sum_{t \in T} \sum_{j \in J} \Delta_t (\lambda_{DGj} - c_{DGj}) P_{DGj}(t) \quad (1)$$

subject to:

$$\sum_{n \in N} \sum_{j \in J} \mu_{nj} = N_{DG}; \quad \mu_{nj} \in \{0, 1\} \quad (2)$$

$$\sum_{n \in N} \mu_{nj} \leq 1 \quad (3)$$

$$\text{Min}_{P_{DGj}(t), P_{SE}(t)} \sum_{t \in T} \Delta_t \rho_{SE}(t) P_{SE}(t) + \sum_{t \in T} \sum_{j \in J} \Delta_t \lambda_{DGj} P_{DGj}(t) \quad (4)$$

subject to:

$$P_{SE}(t) - P_{Dn}(t) - P_n(t) = 0; \quad \forall n \in N / n=SE, \forall t \in T \quad (5)$$

$$Q_{SE}(t) - Q_{Dn}(t) - Q_n(t) = 0; \quad \forall n \in N / n=SE, \forall t \in T \quad (6)$$

$$\mu_{nj} P_{DGj}(t) - P_{Dn}(t) - P_n(t) = 0; \quad \forall n \in N / n \neq SE, \forall t \in T \quad (7)$$

$$\mu_{nj} Q_{DGj}(t) - Q_{Dn}(t) - Q_n(t) = 0; \quad \forall n \in N / n \neq SE, \forall t \in T \quad (8)$$

$$P_{SE}^{Min} \leq P_{SE}(t) \leq P_{SE}^{Max}; \quad \forall t \in T \quad (9)$$

$$Q_{SE}^{Min} \leq Q_{SE}(t) \leq Q_{SE}^{Max}; \quad \forall t \in T \quad (10)$$

$$P_{DGj}^{Min} \leq P_{DGj}(t) \leq P_{DGj}^{Max}; \quad \forall j \in J, \forall t \in T \quad (11)$$

$$Q_{DGj}^{Min} \leq Q_{DGj}(t) \leq Q_{DGj}^{Max}; \quad \forall j \in J, \forall t \in T \quad (12)$$

$$V_n^{Min} \leq V_n(t) \leq V_n^{Max}; \quad \forall n \in N, \forall t \in T \quad (13)$$

$$-S_{l_{mn}}^{Max} \leq S_{l_{mn}}(t) \leq S_{l_{mn}}^{Max}; \quad \forall l_{mn} \in L, \forall t \in T \quad (14)$$

The active and reactive power injections are given by following equations.

$$P_n = V_n \sum_{m \in N} V_m [g_{nm} \cos(\theta_{nm}) + b_{nm} \sin(\theta_{nm})] \quad (15)$$

DG1	DG2	...	DGJ	
Bus_DG1	Bus_DG2	...	Bus_DGJ	→ Location
Price_DG1	Price_DG2	...	Price_DGJ	→ Contract Price

Fig. 2. Codification proposal.

$$Q_n = V_n \sum_{m \in N} V_m [g_{nm} \sin(\theta_{nm}) + b_{nm} \cos(\theta_{nm})] \quad (16)$$

The apparent power flow in the line connecting nodes m , n is given by following equations.

$$S_{lmn} = \sqrt{P_{lmn}^2 + Q_{lmn}^2} \quad (17)$$

$$P_{lmn} = V_m^2 g_{mn} - V_m V_n g_{mn} \cos(\theta_{mn}) - V_m V_n b_{mn} \sin(\theta_{mn}) \quad (18)$$

$$Q_{lmn} = -V_m^2 b_{mn} + V_m V_n b_{mn} \cos(\theta_{mn}) - V_m V_n g_{mn} \sin(\theta_{mn}) \quad (19)$$

The above model is solved using a specialized genetic algorithm. Note that in order to evaluate the objective function given by (1) it is necessary to know the location, contract prices and energy sold by the DG units. The first two sets of variables (location and contract prices) are encoded, for each individual candidate, in a two-row array as shown in Fig. 2. Note that the enforcement of constraint (2) is guaranteed by the size of the array; that is, the number of columns must be equal to the number of DG units. To enforce constraint (3) it is verified that all entries of the upper row are different from each other and belong to a previously defined set of candidate buses. With the information provided above, the energy sold by the DG units is calculated by solving the inner optimization problem which consists on an optimal dispatch given by (4)–(14). Once the inner optimization problem is solved, the value of the upper level objective function can be calculated.

The GA operates considering an initial population of candidate solutions encoded as described above. A series of operations, consisting in selection, recombination, mutation and local improvement are successively applied over the initial population until a stopping criterion is met. The GA aims at the improvement of the objective function at every iteration, preserving the best individuals and the diversification of the population. The objective function of the GA is the profit maximization of the DG owner; however, this profit maximization is subject to the minimization of payments procured by the DisCo (solved as an optimal dispatch problem given by (4)–(14)).

The specialized GA implemented in this paper has two main characteristics that differentiate it from conventional GAs: (i) a substitution logic that preserves the best individuals and avoids premature convergence, and (ii) a local improvement strategy that performs an exhaustive search for better individuals within a predefined neighborhood. Conventional GAs lack these two characteristics, and, therefore, (for this particular problem) are likely to find lower-quality solutions. Several tests performed by the authors showed that suppressing the local improvement stage results in less computational time, at the expense of finding lower-quality solutions. On the other hand, changing the substitution criteria for that of a conventional GA led, in most cases, to premature convergence. A detailed explanation of the GA implemented in this work is provided in Section 3.

3. Specialized genetic algorithm

A GA can be considered as a technique of combinatorial optimization that is suitable for approaching multimodal and/or non-differentiable problems. This type of algorithms, initially proposed by

Holland in [23], are based on the principle of natural selection in which the fittest individuals have more chances to survive and transmit the genetic code to their offspring. In a GA the candidate solutions of the problem are encoded in a string or chromosome. The process usually starts from a population of randomly generated individuals. In each generation, the fitness of every individual is evaluated, then some individuals are chosen (based on their fitness) to be recombined and mutated in order to generate the new individuals of the next generation. The process stops after a maximum number of generations have been calculated or when a satisfactory fitness level has been reached.

In this paper a modified version of the GA proposed by Chu and Beasley in [24] is used, being the main difference the codification of the problem that avoids non feasible solutions. The Chu–Beasley genetic algorithm (CBGA) has some special characteristics that differentiate it from conventional GAs. These characteristics are: (i) the use of a fitness and unfitness function to evaluate the objective function and the degree of unfeasibility of an individual, respectively, (ii) the substitution of only one individual in each iteration, and (iii) a local improvement stage for each individual tested. The CBGA was initially designed to solve the generalized assignment problem [24]. However, it has been applied in other fields, such as transmission planning [25]. The flowchart of the CBGA is shown in Fig. 4.

3.1. Problem codification

The coding of the potential solutions is a key aspect in the implementation of a GA. The codification shows how a candidate solution is represented, and it can facilitate or complicate the implementation of the different phases of the GA. The codification proposed in this paper is shown in Fig. 2. In this case, an individual is represented by a two-row array. The upper row indicates the bus in which a DG unit is located and the lower row indicates its contract price offer. The main advantage of this codification is that it prevents the creation of unfeasible individuals in the recombination stage. Due to the fact that the coding of the CBGA presented in this paper does not generate unfeasible solutions, the main differences with conventional GAs are the substitution of a single individual at each iteration and the local improvement stage.

3.2. Initial population

The initial population is randomly generated and limited by Eq. (2). However, it is possible to restrain the location of the DG units to a specific set of buses that represent the preference of the DG owner. The set of candidate buses can also be restricted using any other criterion such as the ones with the highest demand, or those furthest from the substation. As regards to contract price offers, they are discretized bounded by minimum and maximum values. Sensible minimum and maximum values are chosen bearing in mind the wholesale energy market prices and a reasonable profit margin.

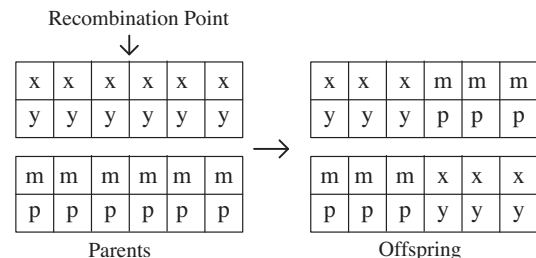


Fig. 3. Example of recombination.

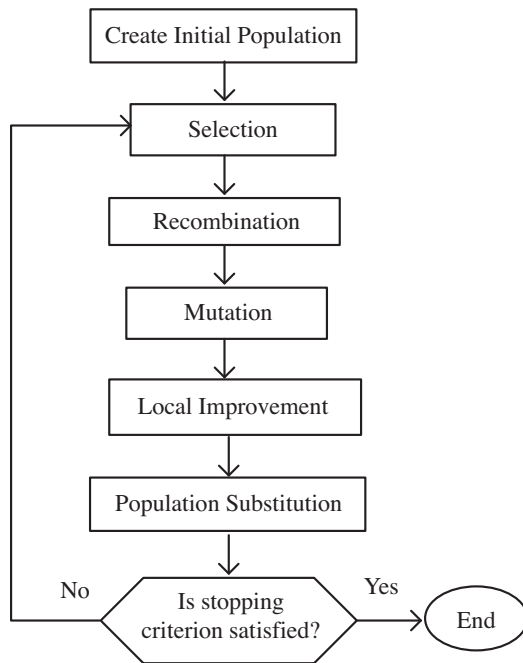


Fig. 4. Flowchart of the CBGA.

3.3. Fitness evaluation

The fitness corresponds to the profits of the DG owner given by (1). The profits are equal to the difference between the operation cost and the contract price multiplied by the energy sold. To calculate the energy sold, it is necessary to solve the optimal dispatch given by (4)–(14). Such dispatch is solved for each individual of the population.

The original version of the CBGA includes an unfitness function that determines the degree of unfeasibility of an individual. Due to the fact that all individuals represented by the proposed codification are feasible, such unfitness function is not necessary.

3.4. Selection

The selection procedure uses a competition based on two tournaments. In each tournament k individuals are randomly selected from the current population and the one with the highest fitness function is selected to be a parent. The two parents selected from the tournaments must be different from each other. After this, both parents go on to the recombination stage.

3.5. Recombination

In the proposed GA a single-point recombination scheme is implemented. In the recombination stage a point of the chromosome is randomly selected, and both parents interchange their information creating two offspring as shown in Fig. 3. In the traditional GA the two offspring can be part of the individuals in the next generation; however, in the CBGA one offspring is randomly selected as candidate to substitute an individual of the current population.

3.6. Mutation

The offspring selected from the recombination stage has to go through the mutation stage. In this case, a position of the chromosome is randomly selected with a given probability and mutated. The mutation consists in changing the information of that particu-

lar position of the chromosome. In this case, the mutation is implemented either on the first or second row with the same probability. If it is done on the first row, then, the location of a particular DG unit is modified moving it to a different bus within a specified range. If the mutation is done on the second row, then the contract price of this DG unit is modified within a specified range.

3.7. Local improvement

One of the most important features of the GA presented in this paper is the local improvement. At this stage, a local search of better individuals is performed. The improvement stage is carried out either on the first or second row with the same probability. If it is done on the first row, then, the location neighborhood of every DG unit is explored, within a specified range, in search of better individuals. If it is done on the second row, the same procedure is performed over the contract price offers. If a better solution proposal is found at this stage, it replaces the current offspring. This local improvement phase permits a more efficient evolution of the GA, and, along with the mutation phase, helps the GA to escape from local optimal solutions.

3.8. Population substitution

Unlike traditional GAs, the CBGA substitutes only one individual in each generation. Such substitution is performed taking into account the quality and feasibility of the candidate individual. In this case, all individuals represented with the proposed codification are feasible. Accordingly, the offspring only replaces the worst individual of the current population if it is better and also different from this one. Otherwise, the current offspring is discharged. This substitution logic preserves the best individuals and the diversification of the population. Furthermore, it avoids premature convergence and facilitates the implementation of local improvement.

The process stops if the incumbent (the best solution found in the process) does not change after a predefined number of generations, or when the maximum number of generations has been reached.

4. Tests and results

To show the effectiveness of the proposed approach several tests are carried out with two distribution systems of 34 and 85 buses, respectively. Fig. 5 depicts a distribution system of 34 buses. This system is similar in topology to the IEEE 34-bus distribution system; however, a single-phase modified version has been used. The line data are provided in the Appendix A. The load distribution in all nodes has been kept proportional to that in the original IEEE 34-bus test system [26]. Fig. 6 shows the load distribution considering a total demand of 10 MW. It can be observed that most of the load is concentrated far from the substation.

4.1. Scenarios under study

The proposed approach is tested to find the location and contract prices of several DG units considering a 1-year contract. For the sake of simplicity and without loss of generality all DG units are considered to have a capacity of 1.5 MW and an operation cost of 55 €/MW h. Furthermore, in this first example, we consider only active power generation of the DG units. Three different scenarios of low, medium and high load have been considered. The annual load duration curves for these scenarios are depicted in Fig. 7. The wholesale energy market prices corresponding to these load scenarios are shown in Fig. 8. We work under the assumption that these prices are not altered due to the activity of DG. Note that Figs.

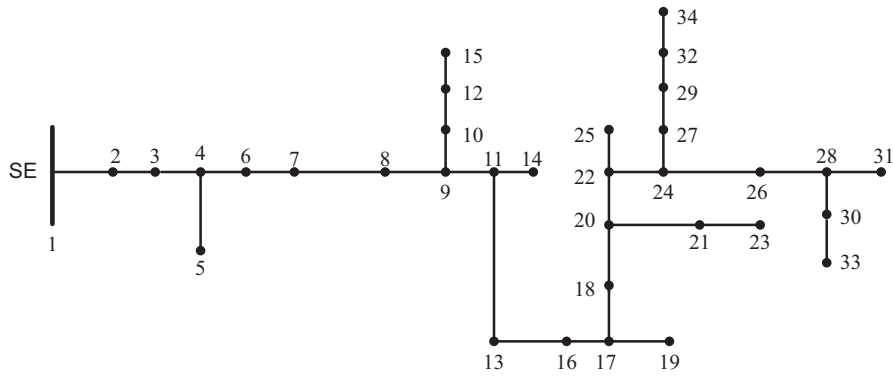


Fig. 5. 34-bus distribution system.

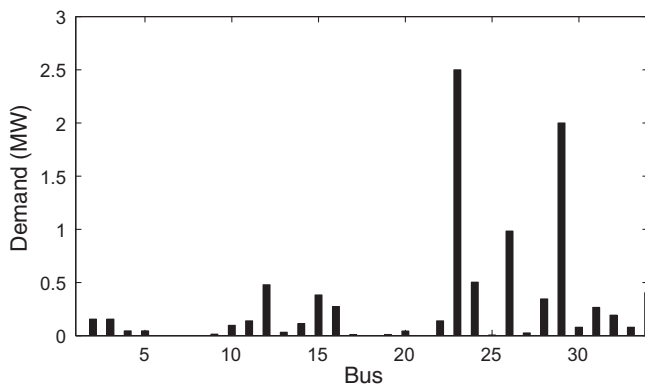


Fig. 6. Load distribution of the 34 bus distribution system for a total demand of 10 MW.

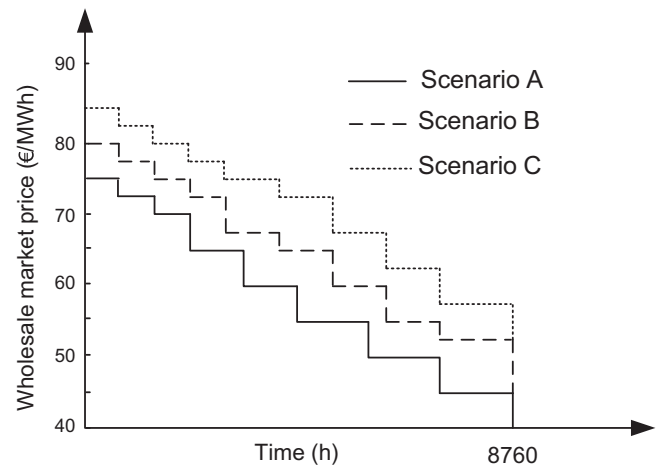


Fig. 8. Wholesale energy market prices for different scenarios.

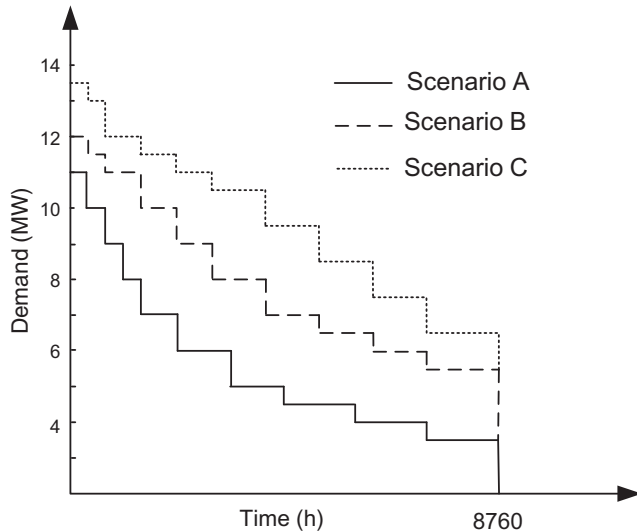


Fig. 7. Annual load duration curves for different scenarios.

7 and 8 exhibit the same shape. That is because high prices in the wholesale energy market are expected to take place precisely during high load periods and vice versa.

4.2. Solution by exhaustive enumeration

The location and contract pricing of the different DG units proposed in this paper represents a combinatorial optimization prob-

lem and, as such, it is solvable by exhaustive enumeration. Nevertheless, even for a reduced number of DG units with a few discrete contract price offers, the number of combinations to be tested can be prohibitive. Due to the fact that the number of candidate solutions increases exponentially with the number of DG units, exhaustive enumeration is performed only with two DG units named DG1 and DG2. These tests are carried out for illustrative purposes and to compare the results with those obtained with the proposed approach based on a GA.

Fig. 9 shows the total profits in scenario C for a contract price offer of 80 €/MWh, considering all possible DG locations. It can be observed that the highest profits are obtained when both DG units are located in the last buses that are far from the substation. Most of the load is concentrated at these buses, as shown in Fig. 6. Since the contract price offer has been kept fixed, the variation of profits is due to the amount of energy purchased by the DisCo. Accordingly, we conclude that the DisCo purchases more energy from the DG units if they are located in the zone with the highest demand.

Fig. 10 shows the total energy losses of the system for different locations of the DG units considering the same contract price offer of 80 €/MWh in scenario C. It can be observed that the highest reduction in energy losses takes place when both DG units are located far from the substation. It can be noted in Figs. 9 and 10 that the locations of the DG units that bring the highest profits to the DG owner are those that result in the highest reduction in power losses for the DisCo. This mutual benefit is due to the fact that the interests of both agents are considered in the optimization process.

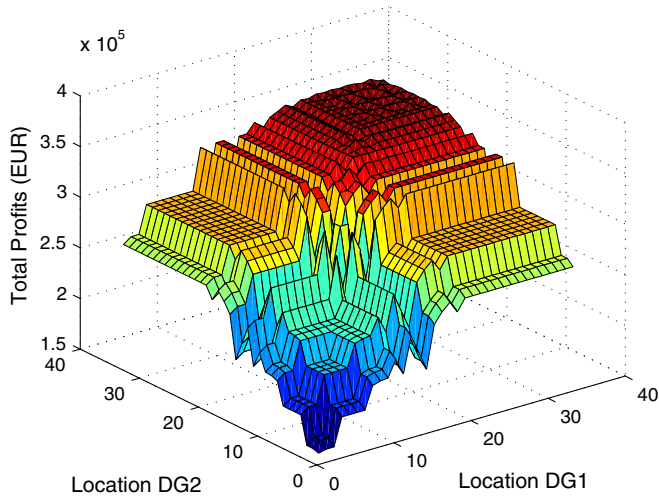


Fig. 9. Total profits (€) for different locations of DG1 and DG2 considering a fixed contract price offer of 80 €/MW h.

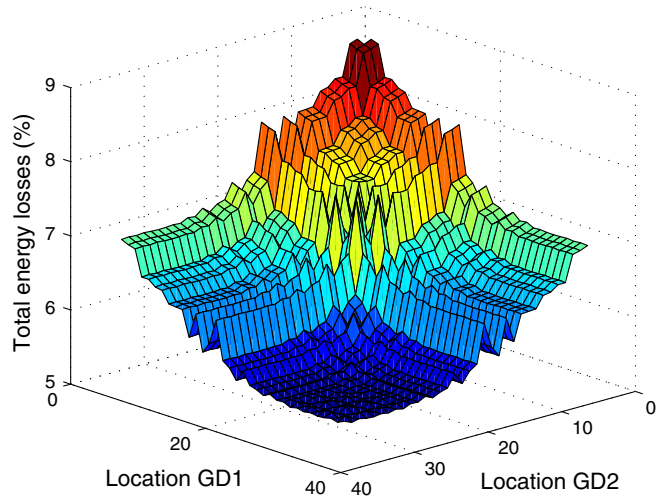


Fig. 10. Total power losses for different locations of DG1 and DG2 considering a fixed contract price offer of 80 €/MW h.

Table 1
Optimal locations and contract prices (in €/MW h) for two DG units using exhaustive enumeration.

DG Unit	Scenario A		Scenario B		Scenario C	
	Bus	CP	Bus	CP	Bus	CP
DG1	29	69	20	70	26	81
DG2	30	73	29	78	29	83
Profits (€)	124698.71		212704.54		367482.09	

Table 1 shows the optimal location and contract prices (CP) for the three scenarios using exhaustive enumeration. The number of candidate solutions to be evaluated is given by:

$$CS = (CB * CPO)^{N_{DG}} \quad (20)$$

where CS is the number of candidate solutions, CB is the number of candidate buses; CPO is the number of contract price offers and N_{DG} is the number of DG units. Note that in order to evaluate the objective function of each candidate solution it is necessary to solve the optimal dispatch given by (4)–(13). This dispatch is solved using a

modified version of the software MATPOWER [27]. To reduce the search space only the last 18 buses of the system are considered as suitable for DG location. Furthermore, for each scenario, only 15 contract price offers are considered and discretized at intervals of 1 €/MW h.

It can be observed in Table 1 that the highest contract prices, as well as the highest profits, are obtained in scenario C, which corresponds to the scenario with the highest demand and wholesale energy market prices. Also, the DG units are located at the last buses, far from the substation, for all scenarios.

4.3. Results using the CBGA

Table 2 shows the locations and contract prices for two DG units obtained with the proposed GA. All tests are performed using a population of 24 individuals with a recombination rate of six individuals and a mutation rate of 10%. The GA is set to stop if the incumbent does not change after 50 iterations or when a maximum of 200 iterations are evaluated. All buses are considered as suitable for DG allocation, and a fine-grained discretization comprising 250 contract price offers at intervals of 0.1 €/MW h is used. Due to this fine discretization, the GA finds slightly better solutions than the exhaustive search for all three scenarios.

Several tests are performed considering 3 and 4 DG units, respectively. However, due to the high number of candidate solutions, the results cannot be compared with an exhaustive enumeration. Fig. 11 shows the convergence of the GA performing three runs for 3 DG units in Scenario C. Despite the fact that the quality of the random-generated initial population is different for every run, the solutions obtained by the GA are quite similar. It can be observed that when the quality of the initial population is poor (third run), the GA needs more iterations to converge.

Table 2
Locations and contract prices (in €/MW h) for two DG units.

DG Unit	Scenario A		Scenario B		Scenario C	
	Bus	CP	Bus	CP	Bus	CP
DG1	29	68.7	27	73.6	24	81.1
DG2	32	73.2	29	78.2	29	82.9
Profits (€)	127135.99		213634.50		367482.31	

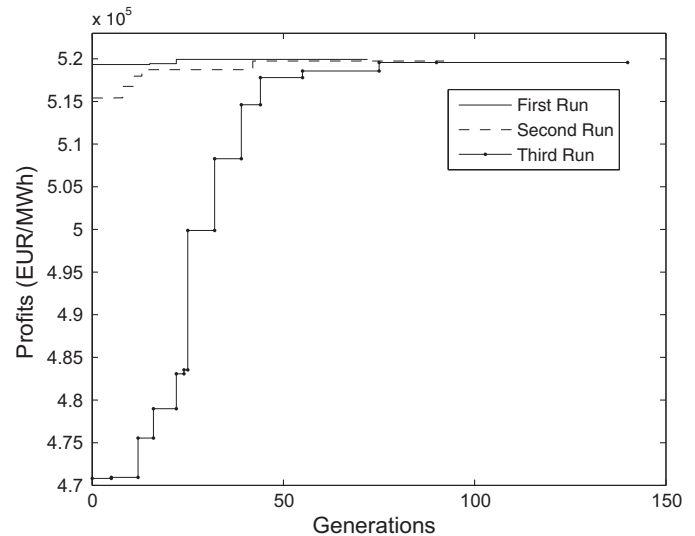


Fig. 11. Convergence of the GA for three different runs considering three DG units in scenario C.

Table 3
Locations and contract prices (in €/MW h) for three DG units.

DG unit	Scenario A		Scenario B		Scenario C	
	Bus	CP	Bus	CP	Bus	CP
DG1	29	71.8	32	76.6	27	79.1
DG2	27	68.7	23	76.4	28	78.9
DG3	23	71.7	24	79.7	20	80.6
Profits (€)	177554.25		301015.50		519933.80	

Table 4
Locations and contract prices (in €/MW h) for four DG units.

DG unit	Scenario A		Scenario B		Scenario C	
	Bus	CP	Bus	CP	Bus	CP
DG1	29	71.1	29	76.6	27	79.1
DG2	18	70.3	13	68.8	30	78.6
DG3	30	70.8	12	70.8	24	78.4
DG4	28	68.6	24	72.4	21	78.1
Profits (€)	222066.00		379969.52		655844.01	

The locations and contract prices obtained for 3 and 4 DG units are shown in Tables 3 and 4, respectively. It can be observed that the contract prices decrease with the introduction of more DG units for all scenarios. Consequently, despite the fact that total profits increase with more DG units, the profits per installed MW diminish, as shown in Fig. 12.

Tables 5 and 6 show the total payments and energy losses of the DisCo for different scenarios, respectively. It can be noted that the DisCo benefits from the presence of DG, since the total payments and the energy losses are reduced with more DG units. It can be observed in Table 6 that when 4 DG units are considered, power losses are reduced over 40% in the three scenarios. At this point it is worth to mention that reduction of power losses is a consequence of the optimization process rather than the objective func-

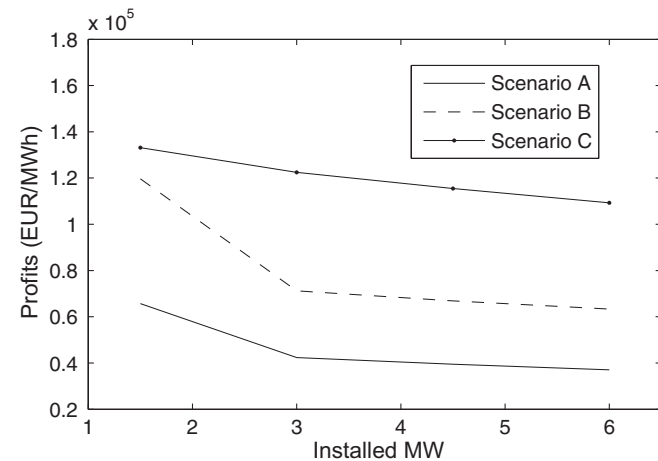


Fig. 12. Profits per installed MW in different scenarios.

Table 5
Total payments of the DisCo for different scenarios.

Number of DG units	Scenario A (€)	Scenario B (€)	Scenario C (€)
0	3189769.93	4979010.96	6706847.78
2	3128060.40	4881089.79	6578757.33
3	3107677.96	4871091.30	6490485.04
4	3083907.87	4769382.91	6423527.83

Table 6
Total energy losses of the DisCo for different scenarios.

Number of DG units	Scenario A (%)	Scenario B (%)	Scenario C (%)
0	5.237	7.199	8.925
2	3.582	5.177	6.354
3	3.239	4.975	5.358
4	2.929	4.232	4.683

tion itself. For high penetration of DG with low contract prices it is likely to find solutions in which power losses increase at the expense of reducing total energy payments. The GA is implemented in MATLAB, using a laptop with a 3 GB RAM and a Dual Core 2.0 GHz processor. The average time for each run of the GA is approx. 3.5 h.

4.4. Tests on an 85-bus distribution system

Fig. 13 depicts an 85-bus distribution system. This system is similar in topology to the one presented in [28]; however, we have modified the line data and demand. The new line data is provided in the Appendix A. Regarding demand, we consider a load duration curve as the one shown in Fig. 7, scenario B, with a lagging power factor of 0.95. For the sake of simplicity the demand is supposed to be equally distributed among the nodes of the network. Furthermore, it is considered that the DisCo can obtain its energy at a fixed price of 65 €/MW h given by a long-term bilateral contract with a retailer. This type of contract hedges the DisCo from the volatility of the wholesale energy market price.

In this case we consider the location and contract pricing of three DG units of 1.5 MW with an operating cost of 55 €/MW h each. Additionally, we suppose that half of the capacity of such units has been already negotiated with a third party or is used for the consumers' own purposes. Such situation is introduced in the model by imposing a lower limit in the power generated by the DG units. We have also included the fact that the DG units can operate with a power factor up to 0.9, either lagging or leading, and, in this way, help out with voltage control. Nevertheless, the pricing of such service is out of the scope of this paper. For the sake of simplicity the inclusion of other voltage control devices is not considered.

Table 7 shows the best solution obtained using the proposed GA. In this case we have considered a population of 12 individuals with a recombination rate of two individuals and a mutation rate of 10%. The GA is set to stop if the incumbent does not change after 50 iterations or when a maximum of 200 iterations are evaluated. All buses are considered as suitable for DG allocation, and the contract price offers are discretized from 60 to 75 €/MW h at intervals of 0.1 €/MW h. The average time for each run of the GA is approx 4.2 h using a laptop with a 3 GB RAM and a Dual Core 2.0 GHz processor.

Despite of the fact that the DisCo can purchase its energy at a fixed price of 65 €/MW h, it is willing to pay up to 65.7 €/MW h to the DG (see Table 7). This is due to the fact that the energy purchased at the substation needs to be transported throughout the distribution system, which increments power losses and deteriorates voltage profile. Consequently, purchasing energy from strategically-located DG units is attractive to the DisCo even when such energy is slightly more expensive than the energy provided through the substation. Table 8 shows the total payments and energy losses of the DisCo with and without DG. In this case we have included different operating modes of DG to illustrate its impact on power losses and payments of the DisCo. It can be observed that the greatest reduction in power losses, as well as in payments, takes place when the DG units supply capacitive reactive power.

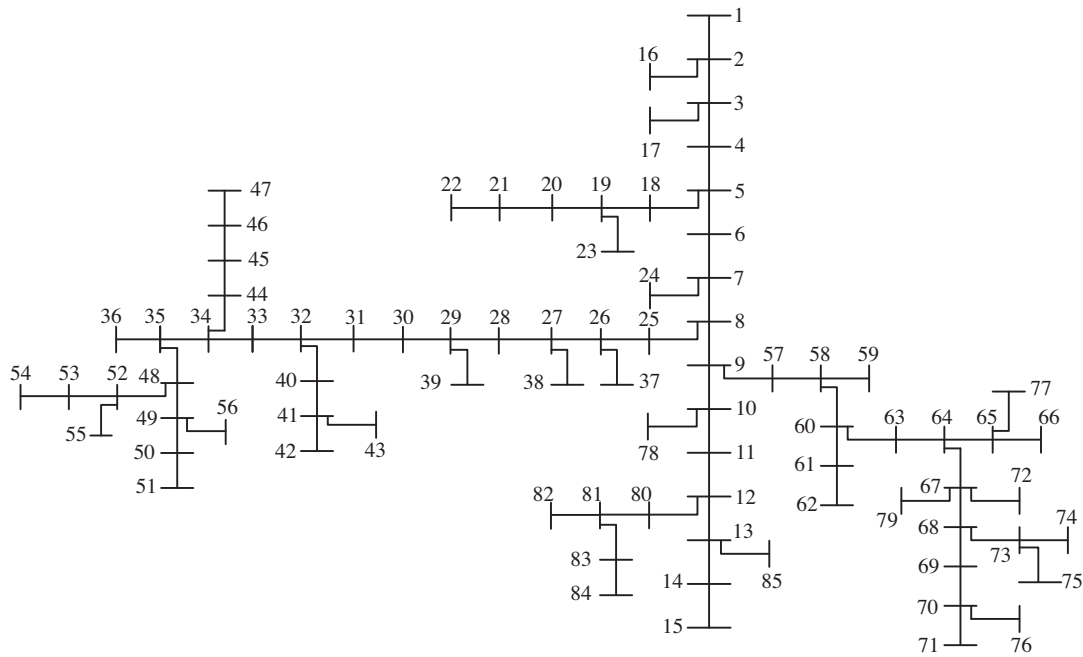


Fig. 13. 85-bus distribution system.

Table 7

Location, contract prices and profits for three DG units.

DG unit	Bus	CP (€/MW h)	Profits (€)
DG1	37	65.6	67593.14
DG2	63	65.7	67665.81
DG3	40	65.6	69642.00

Table 8

Total payments and energy losses of the DisCo.

Case	Total payments (€)	Energy losses (%)
Without DG	4768866.96	7.78
With DG pf = 1.0	4567283.31	2.33
With DG pf = 0.9 leading	4540371.00	1.66
With DG pf = 0.9 lagging	4642861.16	4.02

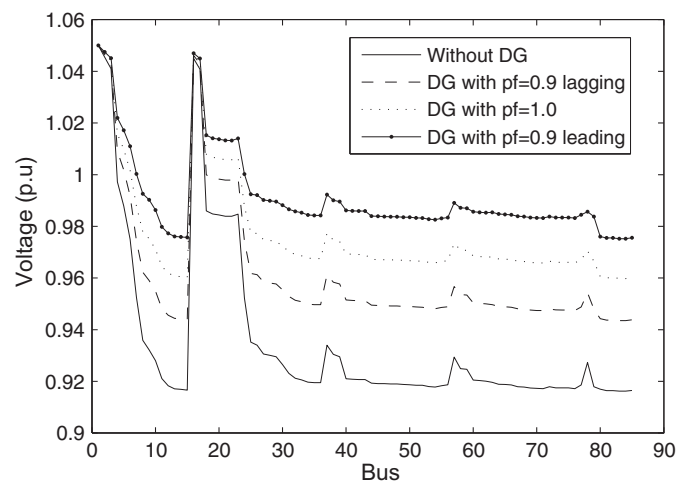


Fig. 14. Voltage profile during peak demand considering different operating modes of DG.

Fig. 14 depicts the impact of DG on voltage profile considering different power factors during peak demand. In this case the three DG units are operating at their maxima. It can be observed that the greatest improvement on voltage profile is achieved when the DG units operate at a leading power factor of 0.9.

5. Conclusions

A specialized GA for the location and contract pricing of DG in distribution networks is presented. The proposed algorithm is envisaged under a nonlinear bilevel programming approach that considers two decision-making agents, namely, the DG owner and the DisCo. The interests of both agents, expressed by two different objective functions, are taken into account, so that the solution of the nonlinear BPP benefits both agents. Due to the fact that under the proposed bilevel programming approach the DisCo must decide over the quantity of energy to be purchased from the DG units, only dispatchable technologies are considered. Further work will include stochasticity in the model along with a more flexible market structure and a more detailed model of the network that allows the consideration of other constraints such as short circuit levels and reliability criteria.

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Appendix A

The line data of the 34-bus and 85-bus distribution systems used in Section 3 are provided in Tables 9 and 10, respectively. Data are given in per unit using a 10 MW base. For the sake of simplicity a line rating of 20 MW is considered for all lines in both distribution systems.

Table 9

Line data of the 34-bus distribution system.

Line	R	X	Line	R	X
1–2	0.0043	0.0040	17–18	0.0128	0.0106
2–3	0.0029	0.0021	18–20	0.0006	0.0006
3–4	0.0280	0.0228	20–21	0.0087	0.0062
4–5	0.0006	0.0005	20–22	0.0117	0.0117
4–6	0.0060	0.0060	21–23	0.0012	0.0006
6–7	0.0016	0.0015	22–25	0.0006	0.0005
7–8	0.0126	0.0094	22–24	0.0012	0.0007
8–9	0.0005	0.0005	24–26	0.0173	0.0107
9–10	0.0122	0.0122	24–27	0.0061	0.0061
9–11	0.0312	0.0283	26–28	0.0006	0.0005
10–12	0.0143	0.0108	28–30	0.0007	0.0006
12–15	0.0157	0.0107	30–33	0.0061	0.0055
11–14	0.0061	0.0062	28–31	0.0116	0.0076
11–13	0.0126	0.0106	27–29	0.0022	0.0022
13–16	0.0028	0.0018	29–32	0.0062	0.0067
16–17	0.0062	0.0061	32–34	0.0078	0.0056
17–19	0.0170	0.0171			

Table 10

Line data of the 85-bus distribution system.

Line	R	X	Line	R	X
1–2	0.0029	0.0020	40–41	0.0030	0.0028
2–3	0.0029	0.0021	41–42	0.0060	0.0050
3–4	0.0281	0.0228	41–43	0.0060	0.0050
4–5	0.0060	0.0050	32–33	0.0016	0.0015
5–6	0.0090	0.0080	33–34	0.0026	0.0043
6–7	0.0160	0.0140	34–44	0.0050	0.0050
7–8	0.0126	0.0093	44–45	0.0022	0.0022
8–9	0.0050	0.0050	45–46	0.0014	0.0016
9–10	0.0173	0.0123	46–47	0.0042	0.0043
10–11	0.0314	0.0284	34–35	0.0007	0.0007
10–78	0.0314	0.0284	35–36	0.0011	0.0012
11–12	0.0143	0.0108	35–48	0.0027	0.0031
12–80	0.0113	0.0184	48–49	0.0028	0.0018
80–81	0.0063	0.0075	49–56	0.0062	0.0062
81–82	0.0013	0.0025	49–50	0.0020	0.0024
81–83	0.0060	0.0080	50–51	0.0078	0.0093
83–84	0.0013	0.0025	48–52	0.0060	0.0075
12–13	0.0158	0.0107	52–55	0.0087	0.0087
13–85	0.0314	0.0284	52–53	0.0117	0.0117
13–14	0.0060	0.0062	53–54	0.0115	0.0060
14–15	0.0127	0.0106	9–57	0.0060	0.0070
2–16	0.0280	0.0180	57–58	0.0115	0.0075
3–17	0.0062	0.0062	58–59	0.0125	0.0107
5–18	0.0170	0.0170	58–60	0.0115	0.0105
18–19	0.0128	0.0106	60–61	0.0060	0.0075
19–23	0.0060	0.0055	61–62	0.0075	0.0060
19–20	0.0087	0.0062	60–63	0.0027	0.0030
20–21	0.0117	0.0117	63–64	0.0022	0.0028
21–22	0.0012	0.0006	64–65	0.0022	0.0022
7–24	0.0060	0.0055	65–77	0.0063	0.0057
8–25	0.0012	0.0007	65–66	0.0078	0.0056
25–26	0.0022	0.0006	64–67	0.0043	0.0040
26–37	0.0012	0.0011	67–79	0.0029	0.0021
26–27	0.0060	0.0050	67–72	0.0030	0.0028
27–38	0.0075	0.0060	67–68	0.0006	0.0005
27–28	0.0012	0.0006	68–73	0.0060	0.0055
28–29	0.0012	0.0009	73–74	0.0016	0.0015
29–39	0.0022	0.0022	73–75	0.0013	0.0009
29–30	0.0063	0.0057	68–69	0.0050	0.0060
30–31	0.0078	0.0056	69–70	0.0022	0.0025
31–32	0.0043	0.0040	70–71	0.0031	0.0034
32–40	0.0029	0.0021	70–76	0.0042	0.0047

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